

7.2 HARDWARE REQUIREMENTS

1. **ESP32 Development Board**

A powerful microcontroller with built-in Wi-Fi and Bluetooth Low Energy (BLE) capabilities. Acts as the BLE receiver at the workplace entrance.

2. **SparkFun NanoBeacon IN100**

Used as the BLE beacon that each employee will carry. Continuously emits a unique UUID to identify the employee.

3. **Coin Cell Batteries (e.g., CR2032)**

Powers the BLE beacons. Chosen for their compact size and long battery life.

4. **USB Cable and Power Adapter**

Required to power the ESP32 board.

5. **Wi-Fi Router / Access Point**

Provides internet connectivity to the ESP32 for transmitting data to Firebase.

6. **Computer or Laptop**

Used for development, programming, and deployment of the system.